

Global Fertilizer Trade Inertia Model

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General Idea

This model estimates post-catastrophe fertilizer availability by assuming trade inertia: in the short term, global trade relationships are “sticky.” Countries attempt to maintain their historical trade partners rather than instantly optimizing for the lowest global cost. The model solves for a new trade matrix that minimizes the deviation from the historical trade structure while strictly satisfying two hard constraints:

1. **Supply Limits:** No country can export more than its remaining production surplus.
2. **Demand Limits:** No country receives more than its remaining demand deficit.

This optimization can be done using the RAS algorithm, which is what this document describes. Note that this should be separately for nitrogen, phosphorus, and potassium.

Notation and Variables

Let N be the set of all countries. For any country $i, j \in N$:

- **Inputs:**
 - P_i : Post-war Production Capacity (tons).
 - C_i : Post-war Domestic Demand (tons).
 - T_{ij}^0 : Historical (Pre-war) trade flow from country i to j .
- **Intermediate Variables:**
 - K_i : Domestic production kept for local use.
 - S_i^* : Target Export Supply (surplus).
 - D_i^* : Target Import Demand (deficit).
 - $\hat{S}_i, \hat{D}_i$: Feasible (Normalized) Targets ensuring global balance.
- **Outputs:**
 - x_{ij} : The new post-war trade flow from i to j .
 - F_i : Final Total Fertilizer Available for consumption in country i .

Phase 1: The “Domestic First” Filter

Countries are assumed to prioritize domestic food security. They only export if production exceeds domestic demand, and only import if production falls short.

For each country i , we calculate the **Net Balance** $B_i = P_i - C_i$.

Case A: Surplus Country ($B_i > 0$)

$$K_i = C_i \tag{1}$$

$$S_i^* = B_i \tag{2}$$

$$D_i^* = 0 \tag{3}$$

Case B: Deficit Country ($B_i \leq 0$)

$$K_i = P_i \quad (4)$$

$$S_i^* = 0 \quad (5)$$

$$D_i^* = |B_i| \quad (6)$$

Phase 2: Global Consistency (Normalization)

The RAS algorithm requires that Total Global Exports equal Total Global Imports. In a catastrophe, these sums will rarely match initially. We normalize the targets to the limiting factor.

Let $S_{total} = \sum_i S_i^*$ and $D_{total} = \sum_i D_i^*$.

Scenario: Global Shortage ($S_{total} < D_{total}$)

Supply is the hard constraint. Demand must be scaled down proportionally.

$$\hat{S}_i = S_i^* \quad (7)$$

$$\hat{D}_i = D_i^* \times \left(\frac{S_{total}}{D_{total}} \right) \quad (8)$$

Scenario: Global Surplus ($S_{total} > D_{total}$)

Demand is the hard constraint. Exports must be scaled down (production cuts/stockpiling).

$$\hat{D}_i = D_i^* \quad (9)$$

$$\hat{S}_i = S_i^* \times \left(\frac{D_{total}}{S_{total}} \right) \quad (10)$$

Phase 3: The RAS Algorithm (Iterative Proportional Fitting)

We seek the matrix $X = [x_{ij}]$ that minimizes the information distance from the historical matrix T^0 subject to the feasible constraints $\hat{S}$ and $\hat{D}$.

The Structural Form

The solution takes the biproportional form:

$$x_{ij} = r_i \cdot T_{ij}^0 \cdot c_j \quad (11)$$

Where r_i is a row-scaling factor (supply push) and c_j is a column-scaling factor (demand pull).

The Iterative Algorithm

We initialize row multipliers $r_i^{(0)} = 1$ and column multipliers $c_j^{(0)} = 1$. We then alternate between two steps until convergence.

Step 3.1: Row Update (Scaling Supply)

For iteration k , we update the row multipliers r_i to ensure the sum of exports from i matches the target $\hat{S}_i$:

$$r_i^{(k)} = r_i^{(k-1)} \times \frac{\hat{S}_i}{\sum_j \left(r_i^{(k-1)} \cdot T_{ij}^0 \cdot c_j^{(k-1)} \right)} \quad (12)$$

Step 3.2: Column Update (Scaling Demand)

We update the column multipliers c_j to ensure the sum of imports to j matches the target $\hat{D}_j$:

$$c_j^{(k)} = c_j^{(k-1)} \times \frac{\hat{D}_j}{\sum_i \left(r_i^{(k)} \cdot T_{ij}^0 \cdot c_j^{(k-1)} \right)} \quad (13)$$

Phase 4: Final Availability

Once the matrix x_{ij} converges, we calculate the total fertilizer available for agriculture in country i . This is the sum of what they kept domestically plus what they secured from the global market.

$$F_i = K_i + \sum_{j \in N} x_{ji} \quad (14)$$

The value F_i represents the final input parameter for the crop simulation models.

Why the RAS Algorithm Works (Entropy Minimization)

The RAS (iterative proportional fitting) updates are not ad hoc: they solve a well-defined convex optimization problem. In particular, they compute the matrix $X = [x_{ij}]$ that matches the feasible row and column totals $\hat{S}$ and $\hat{D}$ while staying as close as possible to the historical pattern T^0 .

The optimization problem

A standard formulation is the (generalized) Kullback–Leibler divergence, also called the *I-divergence*:

$$\min_{x_{ij} \geq 0} \Phi(X) = \sum_i \sum_j \left[x_{ij} \ln \left(\frac{x_{ij}}{T_{ij}^0} \right) - x_{ij} + T_{ij}^0 \right] \quad (15)$$

subject to the hard constraints

$$\sum_j x_{ij} = \hat{S}_i \quad \forall i, \quad (16)$$

$$\sum_i x_{ij} = \hat{D}_j \quad \forall j. \quad (17)$$

The extra terms $-x_{ij} + T_{ij}^0$ do not change the optimizer when the row/column sums are fixed, but they make Φ a true divergence (always ≥ 0 and equal to 0 only when $X = T^0$).

Lagrangian and biproportional form

Form the Lagrangian with multipliers λ_i (rows) and μ_j (columns):

$$\mathcal{L}(x, \lambda, \mu) = \sum_{i,j} \left[x_{ij} \ln \left(\frac{x_{ij}}{T_{ij}^0} \right) - x_{ij} + T_{ij}^0 \right] - \sum_i \lambda_i \left(\sum_j x_{ij} - \hat{S}_i \right) - \sum_j \mu_j \left(\sum_i x_{ij} - \hat{D}_j \right). \quad (18)$$

For $T_{ij}^0 > 0$, the first-order condition is

$$\frac{\partial \mathcal{L}}{\partial x_{ij}} = \ln \left(\frac{x_{ij}}{T_{ij}^0} \right) - \lambda_i - \mu_j = 0, \quad (19)$$

so

$$x_{ij} = T_{ij}^0 e^{\lambda_i} e^{\mu_j}. \quad (20)$$

Defining row/column scaling factors $r_i = e^{\lambda_i}$ and $c_j = e^{\mu_j}$ yields the RAS structural form

$$x_{ij} = r_i T_{ij}^0 c_j. \quad (21)$$

If $T_{ij}^0 = 0$, the divergence is minimized by $x_{ij} = 0$ as well; i.e., the optimizer preserves the *support* of the historical matrix.

Why the iterative updates converge

- **Existence/uniqueness:** $\Phi(X)$ is strictly convex on $x_{ij} > 0$ because $\frac{d^2}{dx^2}(x \ln x) = 1/x > 0$. The feasible set defined by linear equality constraints and $x_{ij} \geq 0$ is convex, so (when it is nonempty and compatible with the support of T^0) there is a unique global minimizer.
- **Alternating minimization:** Each RAS row-scaling step finds the best (minimum-divergence) matrix that matches the row sums while keeping the current column scalings fixed; the column-scaling step does the analogous projection for column sums. These alternating KL-projections monotonically decrease Φ and converge to the unique minimizer, which is exactly the matrix with row sums $\hat{S}$ and column sums $\hat{D}$.